

distance between mean embeddings



No optimization required

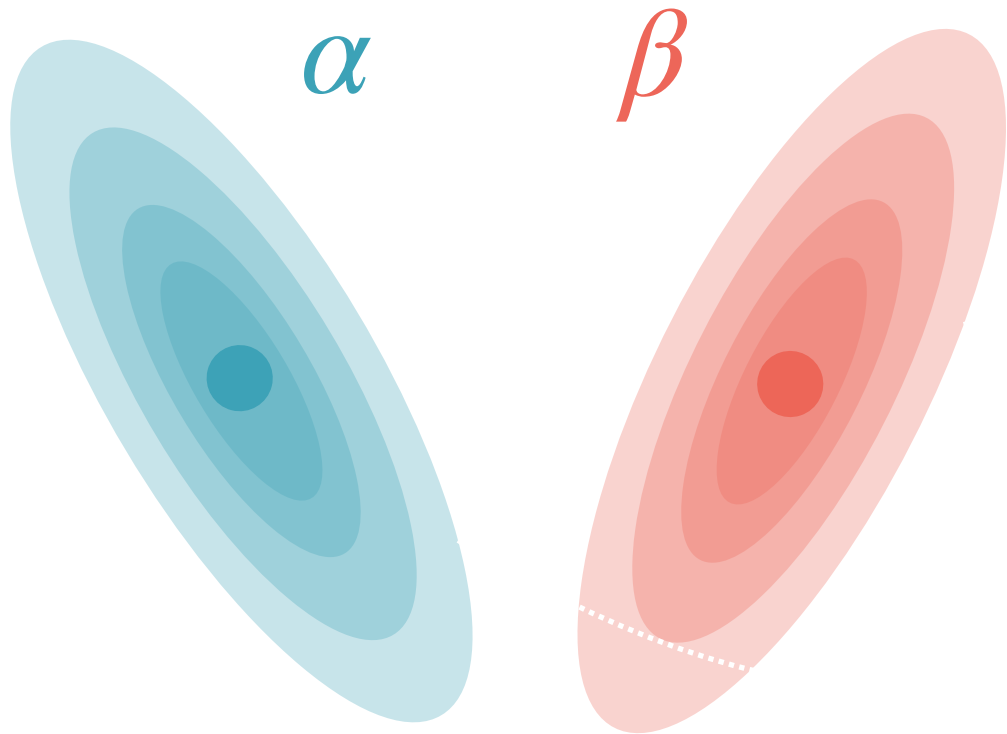


Diagram illustrating the means of the distributions, labeled $\mu_k(\alpha)$ (blue) and $\mu_k(\beta)$ (red). The blue dot represents the mean of the blue distribution, and the red dot represents the mean of the red distribution.

Maximum Mean Discrepancy

- **Definition:** For $\mathcal{H}_k$ Reproducing Kernel Hilbert Space (RKHS) with kernel $k(x, y)$, measures α, β

$$\text{MMD}_k(\alpha, \beta) = \|\mathbb{E}_{x \sim \alpha}[k(x, \cdot)] - \mathbb{E}_{y \sim \beta}[k(y, \cdot)]\|_{\mathcal{H}_k}$$

- **Intuition:**

- Embeds each distribution into a feature space defined by k .
- MMD: **distance between mean embeddings** of the distributions.

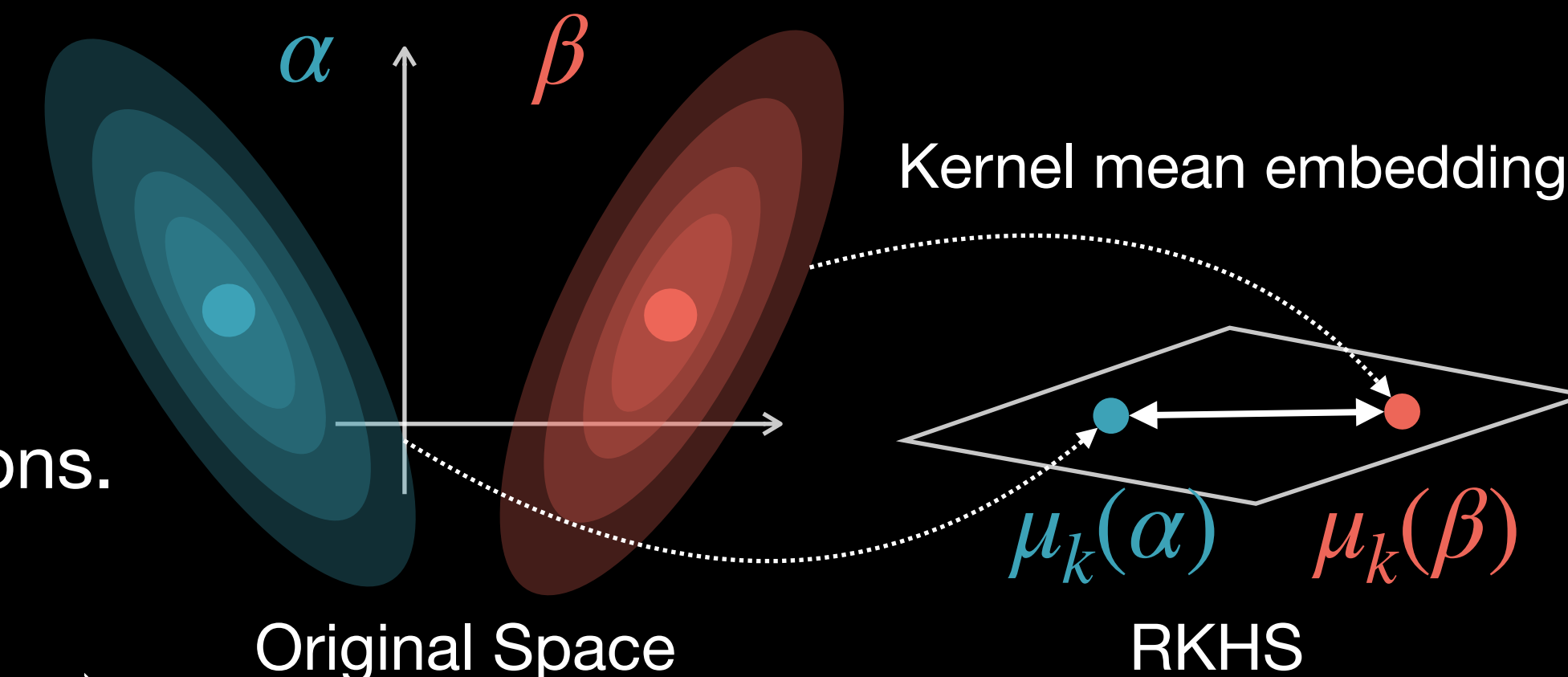
- **Equivalent characterizations:**

$$\text{Variational form: } \text{MMD}_k(\alpha, \beta) = \sup_{\|f\|_{\mathcal{H}_k} \leq 1} \left(\mathbb{E}_{x \sim \alpha}[f(x)] - \mathbb{E}_{y \sim \beta}[f(y)] \right)$$

$$\text{Expectation form: } \text{MMD}_k^2(\alpha, \beta) = \mathbb{E}_{x, x' \sim \alpha}[k(x, x')] + \mathbb{E}_{y, y' \sim \beta}[k(y, y')] - 2 \mathbb{E}_{x \sim \alpha, y \sim \beta}[k(x, y)]$$

- **Properties:**

- $\text{MMD}_k(\alpha, \beta) = 0 \iff \alpha = \beta$ (for characteristic kernels).
- **No optimization required** — just expectations of kernels.
- Often estimated by sample averages, computationally efficient, widely used



SD Interpolates Between OT and MMD

Classic OT

$$\text{OT}(\alpha, \beta) = \min_{\pi \in \Pi(\alpha, \beta)} \int_{X \times Y} c(x, y) \, d\pi(x, y)$$

$\varepsilon \rightarrow 0$

Sinkhorn Divergence

$$S_\varepsilon(\alpha, \beta) = \text{OT}_\varepsilon(\alpha, \beta) - \frac{1}{2}\text{OT}_\varepsilon(\alpha, \alpha) - \frac{1}{2}\text{OT}_\varepsilon(\beta, \beta)$$

$\varepsilon \rightarrow \infty$

Maximum Mean Discrepancy

$$\text{MMD}(\alpha, \beta) = \|\mathbb{E}_{x \sim \alpha}[k(x, \cdot)] - \mathbb{E}_{y \sim \beta}[k(y, \cdot)]\|_{\mathcal{H}_k}$$